



The AI marketplace for reclaimed construction materials

Scan it. Price it. Sell it. Half the cost, a fraction of the carbon.

Team Toronto · Sustainability & Energy track · AI Hackathon · June 2026

Construction waste is money in a landfill

600M t

construction & demolition waste
per year in the EU - the largest
waste stream (~35%)

<10%

finds its way back into a secondary
materials market

€30-80/t

disposal cost paid by site managers
- rising 5-8% per year

Usable steel, brick and timber gets dumped while the project next door buys the same material new at full price.

The seller pays to throw value away; the buyer pays full price and full embodied carbon.

Photo in, priced listing out - AI does the work

1 · Scan

Site manager photographs the leftover pile on the phone

2 · Identify

Grok vision LLM names material, grades condition A/B/C, estimates quantity

3 · Price

Auto-priced at ~50% of new, graded discounts, CO2 saving attached

4 · Match

Buyers browse - or upload a blueprint/BOM and AI matches needs to stock

5 · Transact

Commission on every sale; seller skips disposal cost

AI is the product: vision identification and blueprint matching are what make a fragmented salvage market liquid.

Live demo

Demo 1 - Seller scan

Phone photo of a material pile → AI identifies bricks, rebar, timber → condition grade, quantity, price and CO2 saving → one tap publishes to the marketplace.

Demo 2 - Blueprint match

Upload a blueprint or paste a bill of materials → AI extracts required materials → matched against live used inventory → proposal shows € saved and kg CO2 saved per line.

Screen recording of the working MVP - one responsive web app: phone view scans, desktop view buys.

Both sides save - we take a cut of the win

10%

commission per transaction
(8-12% band)

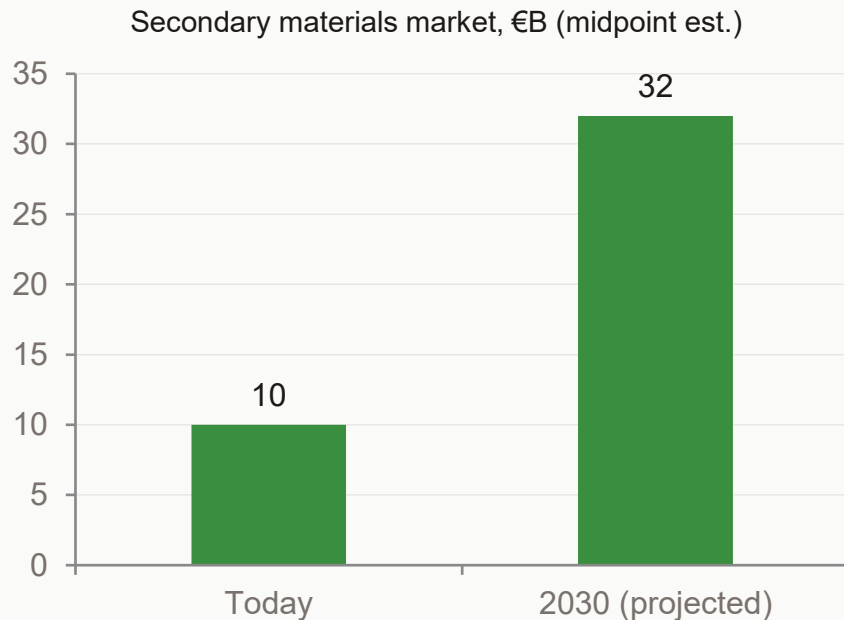
Seller: earns resale value AND avoids €30-80/t disposal

Buyer: pays ~50% of new AND books the carbon saving for ESG reports

Unit economics - one steel beam

New price	€1,000
Listed used (grade A, 50%)	€500
ReBau commission (10%)	€50 revenue
Seller disposal cost avoided	€100-200
Buyer saving vs new	€500 + 1.9 t CO2e/t steel

€100B+ EU market with a regulatory tailwind



- EU construction materials: €150-180B/yr in new sales
- EU Circular Economy Action Plan: 70% construction waste recovery target by 2030 (vs ~30% today)
- Germany pilot region: €8-12B/yr material spend, ~2M t/yr demolition waste around Nuremberg
- ICP: mid-to-large contractors, prefab manufacturers, developers - repeat buyers, high volumes

Why now - the tech and the law just aligned



Vision LLMs got cheap

Material identification from a phone photo is now economically viable - the unlock incumbents lack



ESG reporting is mandatory

EU CSRD pulls scope 3 / embodied carbon into procurement decisions for large firms



Real carbon data exists

EPDs are widespread - CO2 factors per material are spec-grade, not hand-waving



Regulation rewards reuse

German KrWG and EU rules increasingly favour secondary material sourcing

Competition: local salvage yards and manual listing platforms - no AI, no carbon data, no liquidity.

Carbon impact you can put in a report

1 t reused steel $\approx$ 1.9 t CO₂e avoided

Material	Embodied carbon (kg CO ₂ e/kg)
Steel (primary)	1.9
Clay brick	0.24
Concrete	0.13
Timber	stores carbon - reuse keeps it locked in

Example: a 5 t steel spec matched to used stock avoids ~9.5 t CO₂e - in one transaction.

Every purchase ships with auditable scope 3 data buyers can drop straight into ESG reporting.

Factors are computed in the product today - not a roadmap promise.

Next steps - from demo to marketplace

Now	30 days	3-6 months	12 months+
Working MVP: vision scan, auto-pricing, blueprint matching, CO2 engine	Nuremberg pilot: 3-5 site managers, first 10+ live listings, real transactions	Backend + accounts, price feeds for live valuations, logistics partner for pickup	BIM/ERP integrations (Revit, Archicad BOM export), Germany-wide, then EU

Defensibility: inventory data network effects - more listings → better matches → more buyers → more listings.

**Every discarded tonne is revenue
left on the table.**

ReBau recycles it into profit.

The ask: pilot partners in Nuremberg, logistics and ERP intros, API credits to scale the vision pipeline.

Team Toronto · Powered by AI Innovation at FAU Erlangen-Nürnberg